

Surname	Centre Number	Candidate Number
First name(s)		2

**GCE AS/A LEVEL**

2410U10-1

**TUESDAY, 17 MAY 2022 – MORNING****CHEMISTRY – AS unit 1**

**The Language of Chemistry, Structure of Matter
and Simple Reactions**

1 hour 30 minutes

Section A**Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.

Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10(a)**.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 7.	10	
8.	14	
9.	12	
10.	13	
11.	17	
12.	14	
Total	80	

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SECTION AAnswer **all** questions.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in sodium oxide. [2]

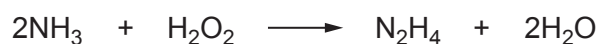
2. Complete the following sentence: [1]

Hydrogen bonding is an intermolecular force that occurs between molecules containing hydrogen atoms bonded to small, atoms which have lone pairs of electrons for example

3. By inserting arrows to represent electrons, show the electronic configuration of a calcium atom. [1]

1s	2s	2p	3s	3p	3d	4s

4. Hydrazine can be manufactured from ammonia.



Calculate the atom economy of this reaction. [1]

atom economy = %



5. The ammonium ion contains a 'coordinate bond'. Explain what is meant by this term. [1]

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.....

6. Uranium is used in nuclear fuel reactors. One of its isotopes, uranium-235, has a half-life of 7.03×10^8 years and decays by α -emission.

- (a) Give the mass number and symbol of the element formed as a product of the radioactive decay of uranium-235. [1]

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- (b) If a quantity of uranium-235 decays, state what fraction is left after 2.812×10^9 years. [1]

fraction left =

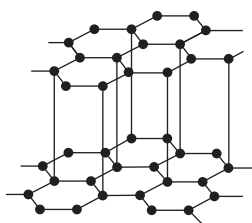
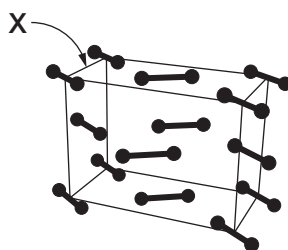
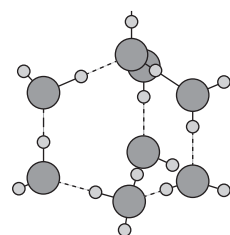
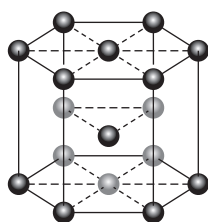
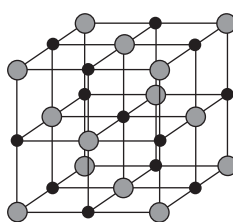
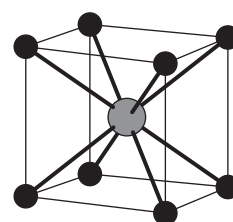
7. Calculate the mass of calcium that contains the same number of atoms as there are molecules in 9.1 g of sulfur dioxide, SO_2 . [2]

calcium mass = g



SECTION BAnswer **all** questions.

8. (a) The diagrams below represent the structures of cadmium metal, caesium chloride, graphite, ice, iodine and sodium chloride.

**A** graphite**B****C****D** cadmium**E****F**

- (i) Label the remaining structures in the spaces provided.

[2]

- (ii) Complete the table showing the trend in melting temperatures of the six substances.

[2]

Substance	Melting temperature
.....	Lowest
.....	
.....	
caesium chloride	
sodium chloride	
.....	
	Highest



- (iii) For structure **B**, name the type of bond or force represented by the letter **X**. [1]

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- (iv) Name the **two** substances which are good electrical conductors in the solid state. [1]

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- (v) Explain why graphite is suitable for use in pencils. [2]

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- (vi) Cadmium is a typical metal.
Give a brief description of metallic bonding. [3]
You may include a diagram to support your answer.

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- (b) Many houses have been built on disused industrial sites. A housing developer wants to test whether the soil on a particular site is contaminated with cadmium ions (Cd^{2+}).

They extract a sample and prepare a solution. Cadmium ions behave in the same way as magnesium ions when treated with sodium hydroxide solution and sodium sulfate solution.

- (i) State what the developer would see if cadmium ions were present when they added:

I. sodium hydroxide solution. [1]

.....

II. sodium sulfate solution. [1]

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- (ii) Write an ionic equation, including state symbols, for **one** of the above observations. [1]

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9. (a) (i) Explain the origins of emission spectra. [2]

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- (ii) Give **one** difference in the appearance of absorption and emission spectra. [1]

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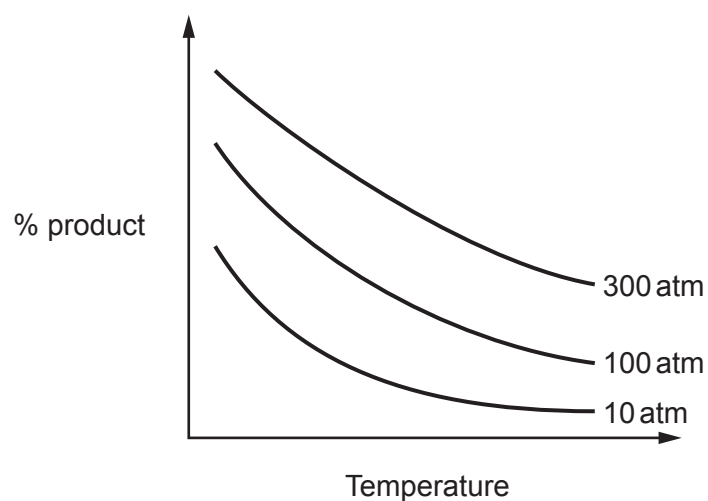
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- (b) A line in the emission spectrum of an element has a wavelength of 95.0 nm. Calculate the frequency of this line in megahertz, MHz. [2]

frequency = MHz



- (c) The diagram below shows how the percentage product varies with temperature and pressure for an equilibrium process.



Use the diagram and Le Chatelier's principle to explain whether:

- (i) the forward reaction is endothermic or exothermic.

[2]

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- (ii) the forward reaction involves an increase or decrease in the number of moles of gas.

[2]

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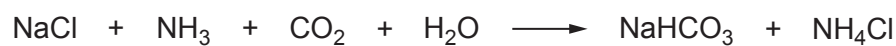
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- (d) The Solvay process is the major industrial process for the manufacture of sodium carbonate. Two of the stages in the process are shown below:



Calculate the maximum mass of sodium carbonate, in kg, which could be obtained from 15.0 tonnes of sodium chloride. Give your answer to an appropriate number of significant figures.

[3]

maximum mass = kg

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10. (a) Group 1 and Group 2 metals are in the s-block of the Periodic Table. Using potassium and calcium as examples, discuss the similarities and differences between Group 1 and Group 2 with respect to:

- the reaction of the metals with cold water.
- the solubility of the carbonates.

[6 QER]

You should include appropriate chemical equations in your answer.

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- (b) The equation for the reaction between potassium carbonate and hydrochloric acid is given below.



A 1.40 g sample of impure potassium carbonate was added to excess dilute hydrochloric acid. The impurity is unreactive and only the potassium carbonate reacts with the acid.

The volume of carbon dioxide released was 186 cm^3 when measured at 298 K and $1.01 \times 10^5\text{ Pa}$.

Calculate the mass of the impurity.

[3]

mass of impurity = g

- (c) A solution is thought to contain potassium chloride. Describe suitable tests that a student could do to confirm this. Include the expected observations.

[2]

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- (d) By referring to ionisation energies, explain why stable compounds containing K^{2+} ions are unlikely to form.

[2]

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11. (a) Arsenic oxide, As_2O_3 , is prepared on an industrial scale by roasting arsenic-containing ores such as arsenopyrite, FeAsS , in air. The other products formed are iron(III) oxide and sulfur dioxide.

(i) State the oxidation state of arsenic in As_2O_3 . [1]

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(ii) Give a balanced chemical equation for the industrial production of As_2O_3 from FeAsS . [2]

.....

- (b) As_2O_3 is moderately soluble in water. 100 cm^3 of a saturated solution at 25°C contains 2.06 g .

When dissolved in water, the oxide reacts to form arsenous acid.



- (i) Calculate the concentration of the arsenous acid, in mol dm^{-3} , in the saturated solution. [3]

concentration of $\text{H}_3\text{AsO}_3 = \dots\dots\dots \text{mol dm}^{-3}$

- (ii) A solution of arsenous acid has a pH of 5.11.

Calculate the hydrogen ion concentration of this solution. [1]

$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$



- (c) The formula for arsenous acid can be written as $\text{As}(\text{OH})_3$ since it contains three hydroxyl (OH) groups bonded to arsenic.

Suggest the shape around the arsenic atom in $\text{As}(\text{OH})_3$. Justify your answer by using VSEPR theory. [3]

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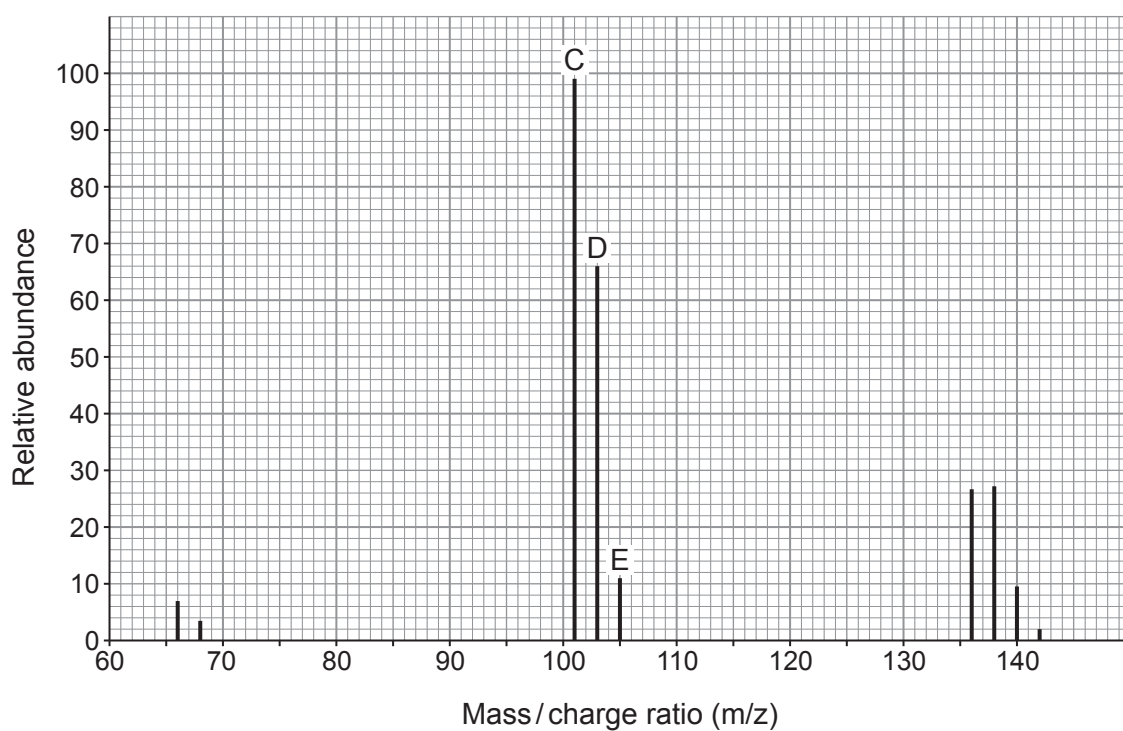
- (d) Phosphorus can form two chlorides, PCl_3 and PCl_5 .

0.181 g of a chloride of phosphorus gave 39 cm^3 of vapour at 1 atm pressure when heated to 87°C . The sample was completely vapourised.

Show that the chloride of phosphorus was PCl_3 . [4]



- (e) The molecular ion region of the mass spectrum of PCl_3 , is shown below.



- (i) Identify the species responsible for peak C at m/z 101. [1]

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- (ii) Explain why the height ratio of peaks C:E is 9:1. [2]

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12. A student was asked to find the identity of a Group 1 metal carbonate by titration.

He was told to use the following method.

- Weigh a sample of the carbonate in a weighing bottle.
- Transfer the carbonate into a beaker and weigh the bottle afterwards.
- Add water to the beaker to dissolve the carbonate.
- Transfer the solution to a volumetric flask.
- Add more water to make the final volume 250.0 cm^3 of solution.
- Accurately transfer 25.0 cm^3 of this solution into a conical flask.
- Add 2–3 drops of a suitable indicator to this solution.
- Fill a burette with 0.100 mol dm^{-3} hydrochloric acid solution.
- Carry out a rough titration of the carbonate solution with the hydrochloric acid.
- Accurately repeat the titration until you get concordant titres and calculate a mean titre.

(a) Another student said that there were two errors in making the 250.0 cm^3 carbonate solution.

Error 1: A small amount of solid remained in the weighing bottle.

Error 2: A small amount of solution remained in the beaker.

Comment on the suggested errors.

If the student is correct suggest how the method could be improved.

If the student is incorrect, explain why.

[2]

Error 1

.....
.....

Error 2

.....
.....

(b) State why he adds an indicator to this solution.

[1]

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.....

(c) Suggest why he was told to carry out a rough titration first.

[1]

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.....



(d) State what you understand by the term 'concordant titres'.

[1]

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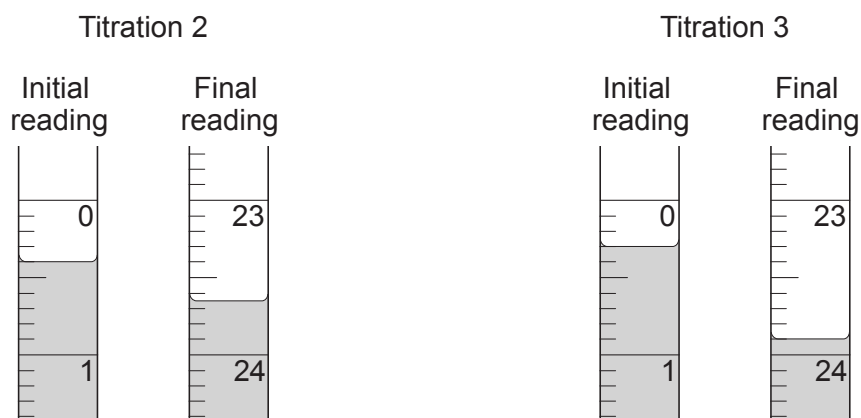
(e) Some of the student's results are shown below:

Mass of weighing bottle + carbonate /g	13.73
Mass of weighing bottle /g	12.48

Titration	Rough	1	2	3
Final reading / cm ³	24.20	23.70		
Initial reading / cm ³	0.00	0.10		
Titre / cm ³				

concentration of hydrochloric acid = 0.100 mol dm⁻³

The diagrams below show the initial burette reading and the final burette reading for the second and third titrations.



Complete the student's table and calculate the mean titre.

[3]

mean titre = cm³



- (f) The equation for the reaction between the metal carbonate and hydrochloric acid is given below. **M** represents the symbol of the Group 1 metal.



- (i) Calculate the number of moles of M_2CO_3 in 25.0 cm^3 of the solution. [2]

number of moles =

- (ii) Calculate the relative formula mass of the carbonate and hence deduce the Group 1 metal in the carbonate.

You **must** show your working. [4]

group 1 metal =

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Examiner
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